

Azure Governance and Resource Organization in Microsoft Azure

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Institution/Organization:

TEESAS - 3MTT NextGen Cohort

Course/Program:

Cloud Computing and Microsoft Azure Fundamentals

Project Description:

This project demonstrates the implementation of Azure governance principles through the use of Subscriptions, Resource Groups, Role-Based Access Control (RBAC), Azure Policy, tagging strategies, naming conventions, and cost management tools within Microsoft Azure.

AZURE GOVERNANCE AND RESOURCE ORGANIZATION PROJECT

Introduction

Cloud governance refers to the processes, policies, and controls used to manage cloud resources effectively. Microsoft Azure provides governance tools that help organizations control costs, manage access, enforce standards, and organize resources efficiently.

This project demonstrates the implementation of Azure governance principles through the use of subscriptions, resource groups, role-based access control (RBAC), Azure Policy, tagging, naming conventions, and cost management.

Project Objectives

The objectives of this project are to:

- Understand the role of Azure Subscriptions.
- Organize resources using Resource Groups.
- Implement Role-Based Access Control (RBAC).
- Apply Azure Policy for governance.
- Implement resource tagging strategies.
- Apply naming conventions for resources.
- Monitor and control cloud costs using budgets and alerts.

Azure Hierarchy

Azure resources are organized in a hierarchical structure as shown below:

Azure Subscription

└─ RG-Networking-Dev

└─ RG-WebApp-Dev

└─ RG-Database-Dev

The subscription serves as the billing and management boundary, while Resource Groups are used to organize related resources for easier administration.

Governance Strategy

Role-Based Access Control (RBAC)

RBAC was used to manage access to Azure resources. The following built-in roles were studied:

- Owner
- Contributor
- Reader

Azure Policy

Azure Policy was used to enforce organizational standards by ensuring resources comply with predefined rules.

Naming Convention

The following naming convention was adopted:

ResourceType-Project-Environment-Number

Examples:

- RG-WebApp-Dev
- VNET-SchoolApp-Dev
- NSG-SchoolApp-Dev
- HybukDB

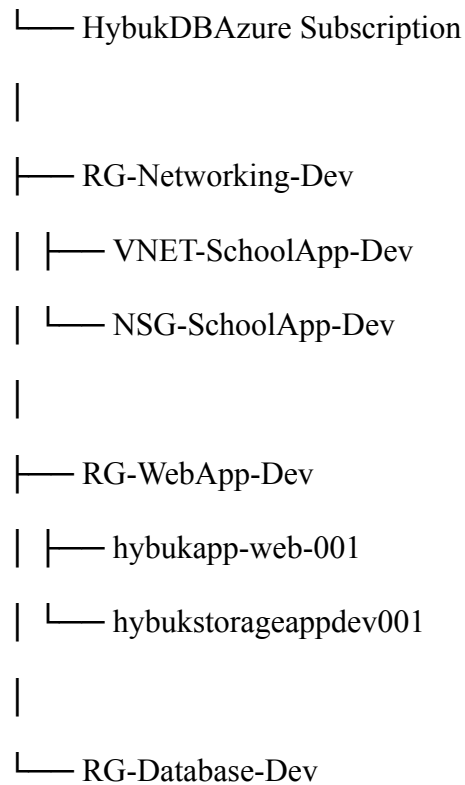
Tagging Strategy

The following tags were applied to resources:

Tag Name	Value
Environment	Development
Department	ICT
Owner	Ibukun

Tags improve resource tracking, governance, and cost management.

Architecture Diagram



Screenshots

Attached is the screenshots of the followings with a brief caption:

1. Azure Subscription Overview
2. Resource Groups Page
3. RBAC (IAM) Configuration
4. Azure Policy Assignment
5. Resource Tags
6. Cost Analysis Dashboard
7. Budget Creation Page
8. Budget Alert Configuration
9. App Service Deployment
10. SQL Database Deployment

Challenges Encountered

1. Some Azure services were not available in the selected region.

Solution: Alternative services and supported regions were selected.

2. Certain resources required globally unique names.

Solution: Unique identifiers were added to resource names.

3. Understanding Azure governance concepts.

Solution: Microsoft Learn documentation and practical exercises were used to improve understanding.

Conclusion

This project provided practical experience in Azure governance and resource organization. Azure Subscriptions, Resource Groups, RBAC, Azure Policy, tagging, naming conventions, and cost management tools were successfully explored and implemented. The project demonstrated how governance practices help improve security, cost control, operational efficiency, and scalability within a cloud environment.